

BAULKHAM HILLS HIGH SCHOOL

YEARLY EXAMINATIONS

YEAR 11

2009

MATHEMATICS

EXTENSION 1

Time Allowed: Two hours
(plus 5 mins reading time)

Instructions:

- Write in the answer booklets provided
- Do not write on this question paper
- Show all working
- Use black or blue pens only
- Start a new page for each question
- Write your name and your teacher's name at the top of each page
- Board approved calculators may be used

Question 1 (12 marks) Start a new page.

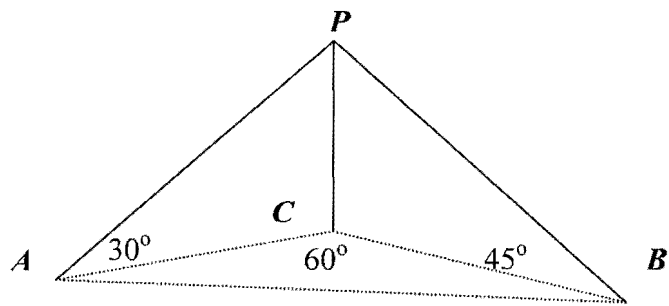
- a) Find the coordinates of the point which divides the interval joining $(-3,1)$ and $(5,6)$ externally in the ratio $3 : 4$ 2
- b) Solve $\frac{4}{x-2} \leq 1$ 3
- c) A curve has parametric equations $x = t + 2$, $y = 4t^2$
Find the Cartesian equation. 2
- d) Evaluate $\lim_{x \rightarrow \infty} \frac{x^2 - 4x + 2}{2x^2 + 3}$ 1
- e) i) Express $\cos x - \sin x$ in the form $R \cos(x + \alpha)$ 2
- ii) Hence solve $\cos x - \sin x = 1$ for $0^\circ \leq x \leq 360^\circ$ 2

Question 2 (12 marks) Start a new page.

- a) Solve the equation $2 \sin^2 \theta = \sin 2\theta$ for $0^\circ \leq \theta \leq 360^\circ$

3
- b) Prove the identity $\frac{2 \sin^3 A + 2 \cos^3 A}{\sin A + \cos A} = 2 - \sin 2A$

2
- c) The angle of elevation of the top of a vertical cliff P from boat A is 30° and from boat B is 45° . The two boats subtend an angle 60° at the base of the cliff and are 1 km apart.



- Find the height of the cliff. (Answer to the nearest metre).

4
- d) For what values of 'b' is the line $y = 3x + b$, a tangent to $y = x^3$?

3

Question 3 (12 marks) Start a new page.

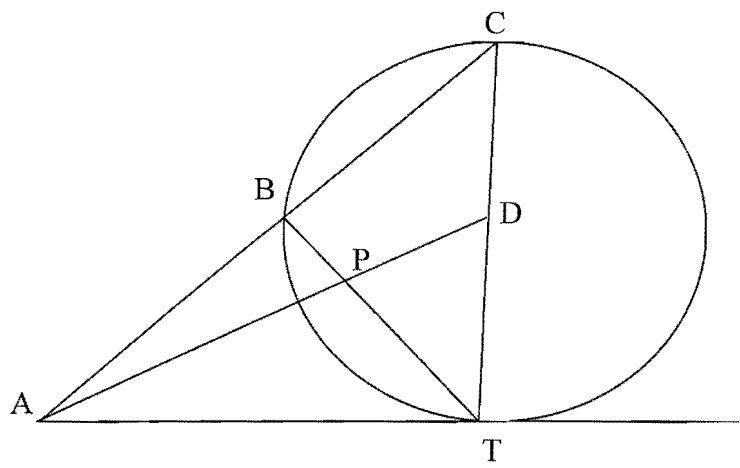
- a) Find the acute angle between lines $2x - 3y + 4 = 0$ and $4x + y - 6 = 0$ to the nearest minute. **3**
- b) Consider the curve $y = \frac{x - 1}{x^2 - 9}$
- i) Find the intercepts with the coordinate axes **2**
- ii) Locate stationary points (if any) **3**
- iii) Write the equations of any asymptotes **2**
- iv) Draw a graph of the function and use it to deduce the number of points of inflexion **2**

Question 4 (11 marks) Start a new page.

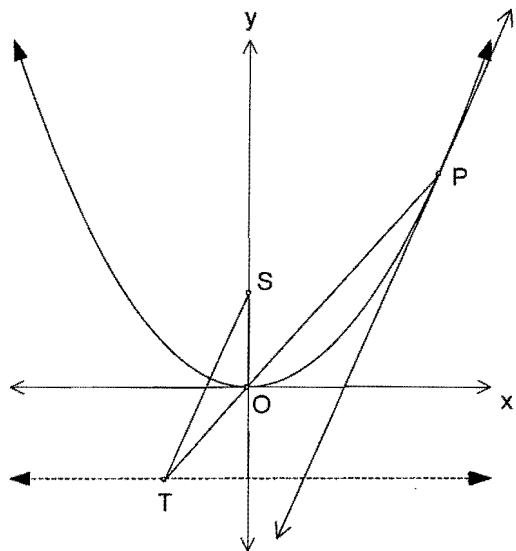
- a) AT is a tangent to the circle at T . ABC is a secant and AD bisects $\angle CAT$

Prove that $TP = TD$

3



- b) The point $P(2ap, ap^2)$ lies on the parabola $x^2 = 4ay$



- i) Derive the equation of the tangent at P
- ii) The line from P to the vertex is produced to cut the directrix at T . If S is the focus, prove that TS is parallel to the tangent at P .

2

3

- c) Differentiate from first principles, $f(x) = \frac{1}{x}$

3

Question 5 (12 marks) Start a new page.

- a) Use mathematical induction to prove that

$$\frac{1}{1 \times 3} + \frac{1}{3 \times 5} + \frac{1}{5 \times 7} + \dots + \frac{1}{(2N-1)(2N+1)} = \frac{N}{2N+1}$$

for positive integers $N \geq 1$

4

- b) If α, β are the roots of $2x^2 - 5x + 8 = 0$
write a quadratic equation with roots α^2, β^2

3

- c) AB is a chord of length 16 units of the circle $x^2 + y^2 = 100$ and P is the mid-point of AB . Determine the locus of P .

2

- d) State the domain of the function $f(x) = \sqrt{9 - x^2} + \frac{1}{x+3}$

3

Question 6 (12 marks) Start a new page.

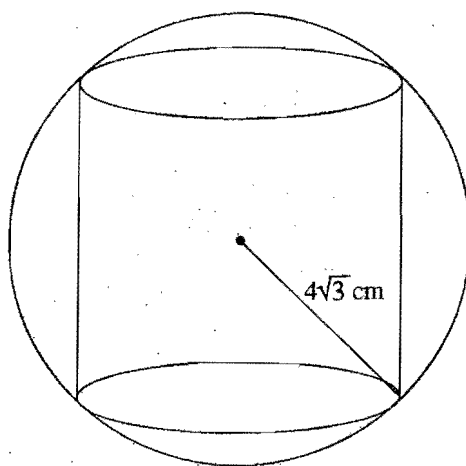
a) Prove that $\tan \theta = \sqrt{\frac{1 - \cos 2\theta}{1 + \cos 2\theta}}$

hence show the exact value of $\tan 22\frac{1}{2}^\circ = \sqrt{2} - 1$

4

- b) Find the height of the cylinder with maximum volume which fits exactly into a sphere of radius $4\sqrt{3}$ cm

4



- c) By making the substitution $t = \tan \frac{\theta}{2}$ or otherwise

show that $\operatorname{cosec} \theta + \cot \theta = \cot \frac{\theta}{2}$

2

d) Solve for x : $2 \cdot 2^x - 5 = \frac{12}{2^x}$

2

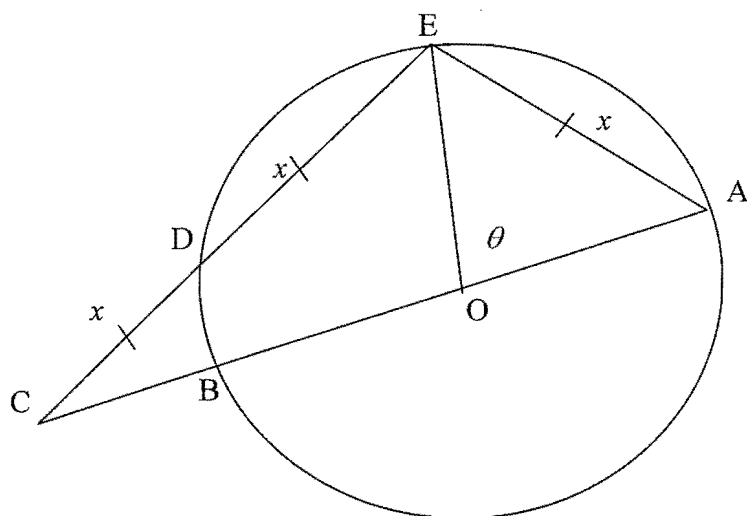
Question 7 (12 marks) Start a new page.

- a) Find the vertex, focus, directrix of parabola with equation

$$(x + 4)^2 = -16(y - 3)$$

3

- b) AB is the diameter of a circle of radius 1 unit with centre O .
 $CD = DE = AE = x$ and $BC = 1$ $\angle AOE = \theta$



Prove $\cos \theta = \frac{1}{4}$

4

- c) If $5 \cos A + 3 = 0$ and $\tan A > 0$ find the exact value of $\cos (A + 90^\circ)$

2

- d) The normals at P and Q on $x^2 = 4ay$ intersect at R whose coordinates are

$$x = a\left(t - \frac{1}{t}\right) \quad y = a\left(t^2 + 1 + \frac{1}{t^2}\right)$$

Find the equation in Cartesian form of the locus of R .

2

- e) Evaluate $\sum_{r=1}^4 r^r$

1

End of Paper

Q1 a) $(-3, 1)$ and $(5, 6)$

-3: 4

$$\left(\frac{-3 \times 5 + 4 \times 6}{-3 + 4}, \frac{-3 \times 6 + 4 \times 1}{-3 + 4} \right)$$

$$= (-27, 14) \quad (1 \text{ each}) \quad [2]$$

b) $\frac{4}{x-2} \leq 1$

$x \neq 2$

$4 = x - 2$

$x = 6 \quad (1)$

Test $x = 3$ false

$\therefore x < 0$ or $x > 6$

c) $x = t + 2, y = 4t^2$

$t = x - 2 \rightarrow y = 4(x-2)^2$

$y = 4(x-2)^2 \quad (1)$

d) $\lim_{x \rightarrow \infty} \frac{x^2 - 4x + 2}{2x^2 + 3} \div x^2$

$\lim_{x \rightarrow \infty} \frac{1 - \frac{4}{x} + \frac{2}{x^2}}{2 + \frac{3}{x^2}}$

$= \frac{1}{2} \quad [1]$

e) i) $\cos(x - \sin x) = \cos(x + \sin x)$

$= 2 \cos x \cos \sin x$

$\therefore \cos x = 1$

$R \sin x = 1$

$\tan x = 1$

$x = 45^\circ, R = \sqrt{2} \quad [2]$

ii) $\cos x - \sin x = \sqrt{2} \cos(x + 45^\circ)$

$\cos x - \sin x = 1$

$\sqrt{2} \cos(x + 45^\circ) = 1$

$x + 45^\circ = 45^\circ, 315^\circ \quad (1) \quad [2]$

$x = 0, 1270^\circ, 360^\circ, 711^\circ$

Q2 a) $2 \sin^2 \theta = \sin 2\theta$

$2 \sin^2 \theta - 2 \sin \theta \cos \theta = 0$

$2 \sin \theta (\sin \theta - \cos \theta) = 0 \quad (1)$

$\therefore \sin \theta = 0$

$\theta = 0, 180, 360^\circ$

$\therefore \theta = 0, 45^\circ, 180^\circ, 225^\circ, 360^\circ$

b) LHS $= 2 \sin^2 A + 2 \cos^2 A$

$= 2(\sin^2 A + \cos^2 A)$

$= 2(\sin^2 A + \cos^2 A)$

$= 2(1 - \sin^2 A)$

$= 2 - 2 \sin^2 A$

$= 2 - \sin^2 A$

$= 2 + \cos^2 A$

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Q2 d) $y = x^3$

$\frac{dy}{dx} = 3x^2$

$y = 3x^2$

$m = 3$

$x = \pm 1$

$x = 1, 3$

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Since $\Delta < 0$

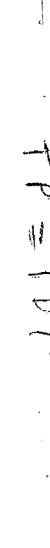
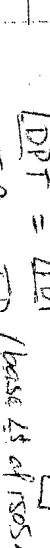
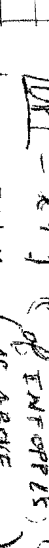
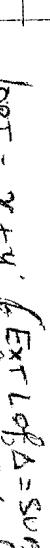
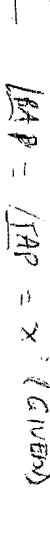
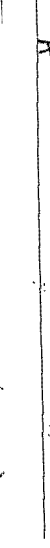
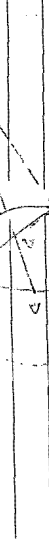
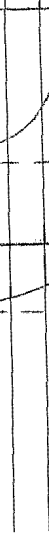
$\therefore$ no stationary points (1)

ii) $x^2 - 9 \neq 0$

$\therefore x \neq \pm 3$

asymptotes at $x = \pm 3$ (1)

$y = 0$ (1)



Since $\Delta < 0$

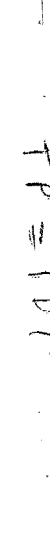
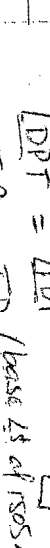
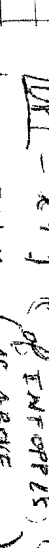
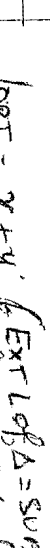
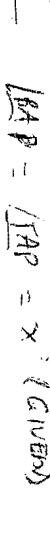
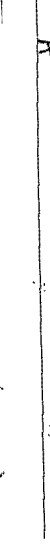
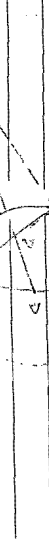
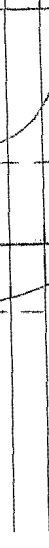
$\therefore$ no stationary points (1)

ii) $x^2 - 9 \neq 0$

$\therefore x \neq \pm 3$

asymptotes at $x = \pm 3$ (1)

$y = 0$ (1)



Since $\Delta < 0$

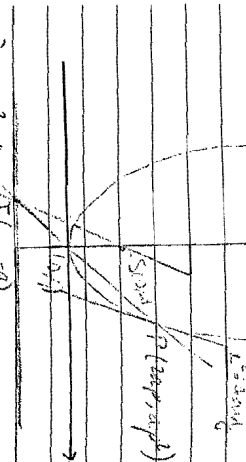
$\therefore$ no stationary points (1)

ii) $x^2 - 9 \neq 0$

$\therefore x \neq \pm 3$

asymptotes at $x = \pm 3$ (1)

Q4 b)



c) $f(x) = \frac{1}{x}$

$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

$= \lim_{h \rightarrow 0} \frac{\frac{1}{x+h} - \frac{1}{x}}{h}$

$= \lim_{h \rightarrow 0} \frac{x - (x+h)}{h(x+h)}$

$= \lim_{h \rightarrow 0} \frac{-h}{h(x+h)}$

Q5 a)

$\frac{1}{1 \times 3} + \frac{1}{3 \times 5} + \frac{1}{5 \times 7} + \dots + \frac{1}{(2n-1)(2n+1)}$

Proof: Let for $n=1$

LHS = $\frac{1}{1 \times 3} = \frac{1}{3}$

RHS = $\frac{1}{2 \times 1 + 1} = \frac{1}{3}$

LHS = RHS

Assume true for $n=k$

ie try to prove

Tangentate for $n=k+1$

LHS = $\frac{1}{1 \times 3} + \frac{1}{3 \times 5} + \dots + \frac{1}{(2k-1)(2k+1)} + \frac{1}{(2k+1)(2k+3)}$

RHS = $\frac{1}{2 \times (k+1) + 1} = \frac{1}{2k+3}$

LHS = $\frac{1}{2k+3} + \frac{1}{2k+3}$

RHS = $\frac{1}{2k+3}$

LHS = RHS

Q5 b) $2x^2 - 5x + 8 = 0$

$x^2 + 3 = \frac{5x}{2}$ $\alpha\beta = 4$

$x^2 + 3 = \frac{5x}{2}$ $= \tan \theta$ (1)

$x^2 + 3 = \frac{5x}{2}$ $= \tan \theta$

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$$66) \cos \theta + \cot \theta = \cot \frac{\theta}{2}$$

$$\text{if } t = \tan \frac{\theta}{2}$$

$$\text{LHS} = \frac{1+t^2}{2t} + \frac{1-t^2}{2t} \quad (1)$$

$$= \frac{1}{t} \quad (1) \quad [2]$$

$$= \frac{1}{\tan \frac{\theta}{2}} = \cot \frac{\theta}{2} = \text{RHS}$$

$$7c) \cos A = -3/5$$

Am QUAD 3

$$\cos(A+90^\circ) = \cos A \cos 90^\circ - \sin A \sin 90^\circ \quad (1)$$

$$[2] = -\frac{4}{5} \cdot 1 = \frac{4}{5} \quad (1)$$

$$d) 2 \cdot 2^x - 5 = \frac{12}{2^x}$$

$$\text{let } 2^x = a$$

$$2a^2 - 5a = 12$$

$$2a^2 - 5a - 12 = 0$$

$$(2a+3)(a-4) = 0 \quad (1) \quad [2]$$

$$2^x = -\frac{3}{2} \text{ or } 2^x = 4$$

$$\text{not possible } x = 2 \quad (1)$$

$$7d) \left(t - \frac{1}{t}\right)^2 = t^2 - 2 + \frac{1}{t^2} \quad (1)$$

$$\left(\frac{7}{a}\right)^2 = \frac{4}{a} - 3 \quad (1)$$

$$7^2/a^2 = \frac{4}{a} - 3 \quad [2]$$

$$x^2 = ay - 3x^2$$

$$67) a) (x+4)^2 = -16(y-3)$$

$$\text{vertex} = (-4, 3) \quad (1)$$

$$16 = 4a$$

$$a = 4$$

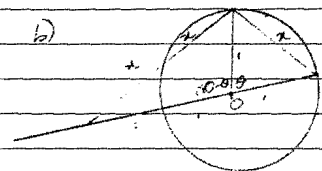
$$\text{focus} = (-4, -1) \quad (1) \quad [3]$$

$$\text{Directrix } y = 7 \quad (1)$$

$$7e) \sum_{r=1}^4 r = 1 + 2 + 3 + 4$$

$$= 1 + 4 + 9 + 16$$

$$= 30 \quad [1]$$



$$\cos \theta = \frac{1^2 + 2^2 - x^2}{2 \times 1 \times 2}$$

$$\cos \theta = \frac{2 - x^2}{2} \quad \text{--- (1)}$$

$$\cos(180^\circ - \theta) = \frac{1^2 + 2^2 - (2x)^2}{2 \times 2 \times 1}$$

$$-\cos \theta = \frac{5 - 4x^2}{4} \quad \text{--- (2) (1)}$$

$$\frac{-2 + x^2}{2} = \frac{5 - 4x^2}{4}$$

$$-4 + 2x^2 = 5 - 4x^2$$

$$x^2 = \frac{9}{6} \therefore x = \frac{3}{2} \quad (1)$$

$$\text{--- (1)} \Rightarrow \cos \theta = \frac{2 - \frac{9}{4}}{2} = \frac{1}{4} \quad (1)$$